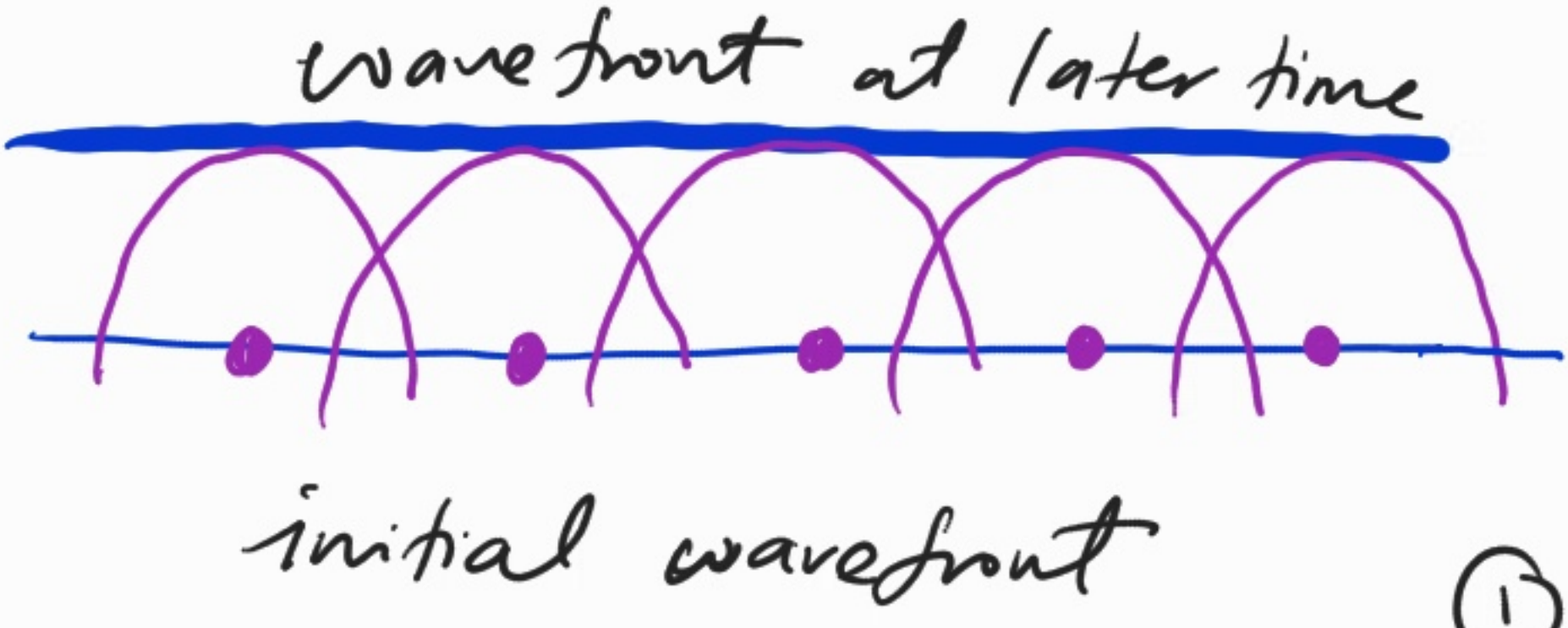
Huygens Principle

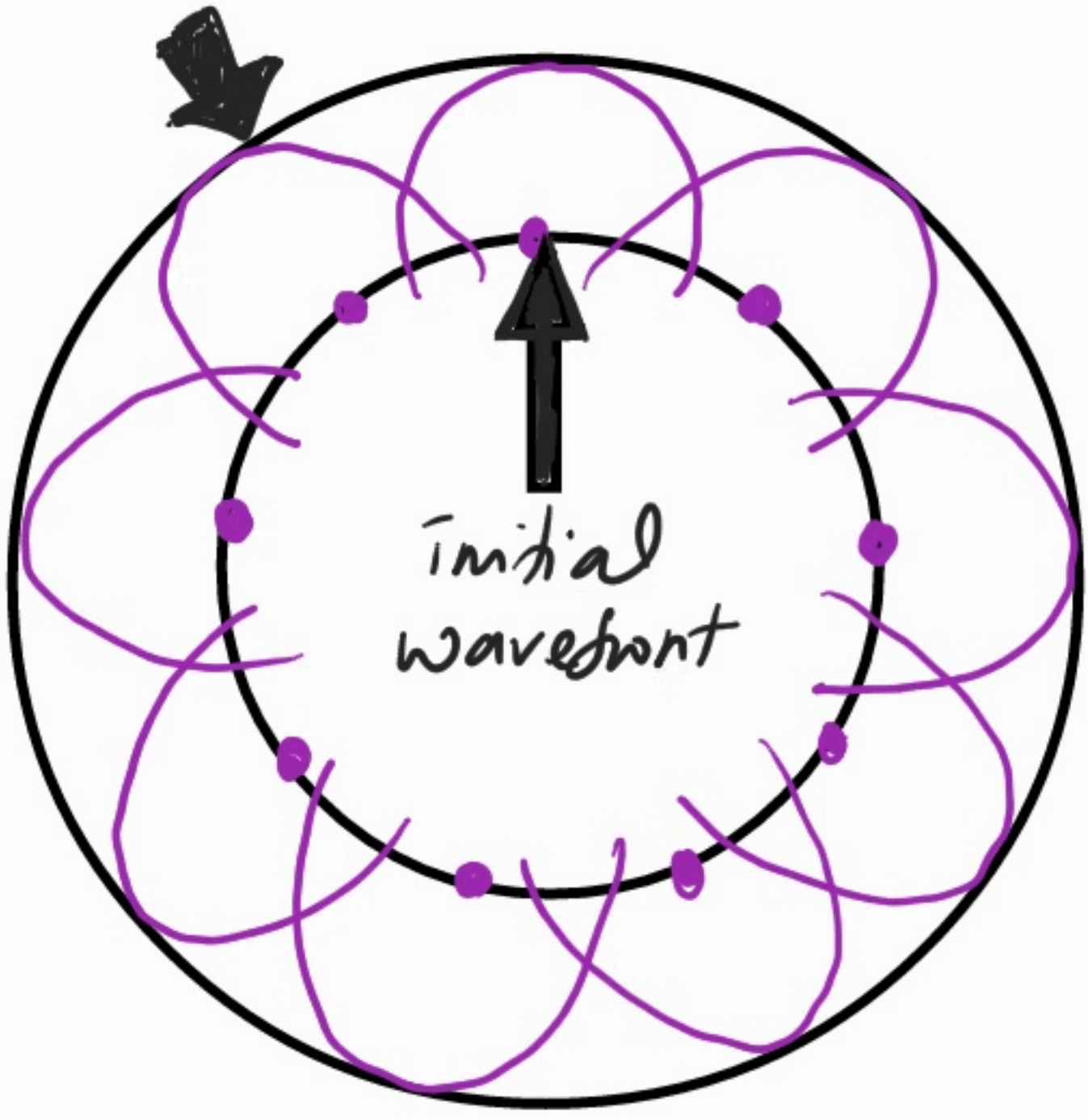
- also called Huygens-Fresnel Principle.
- every point on a wave front is the source of a spherical wave.
- these spherical wavelets mutually interfere — where the sum of these wavelets form a new wavefront.

PLANE WAVE



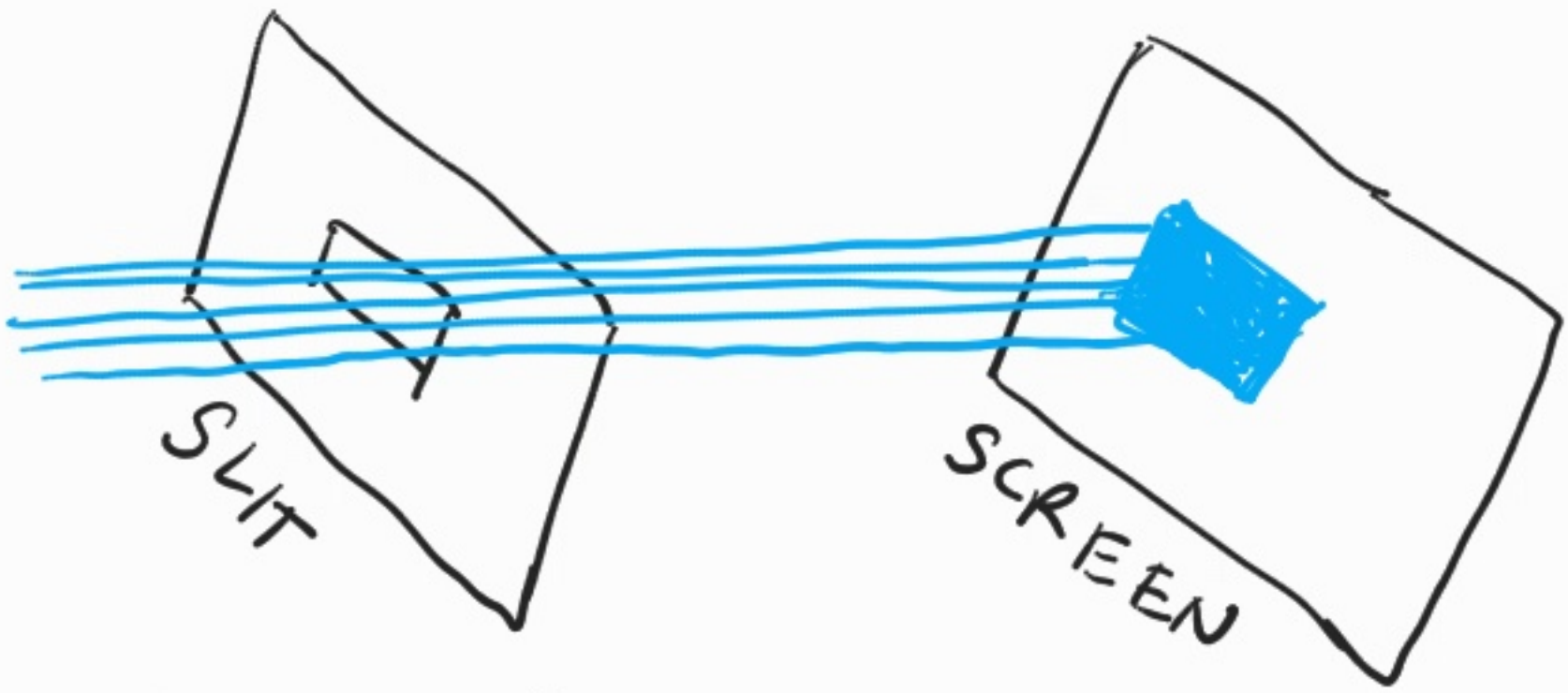
SPHERICAL WAVE

wavefront at later time

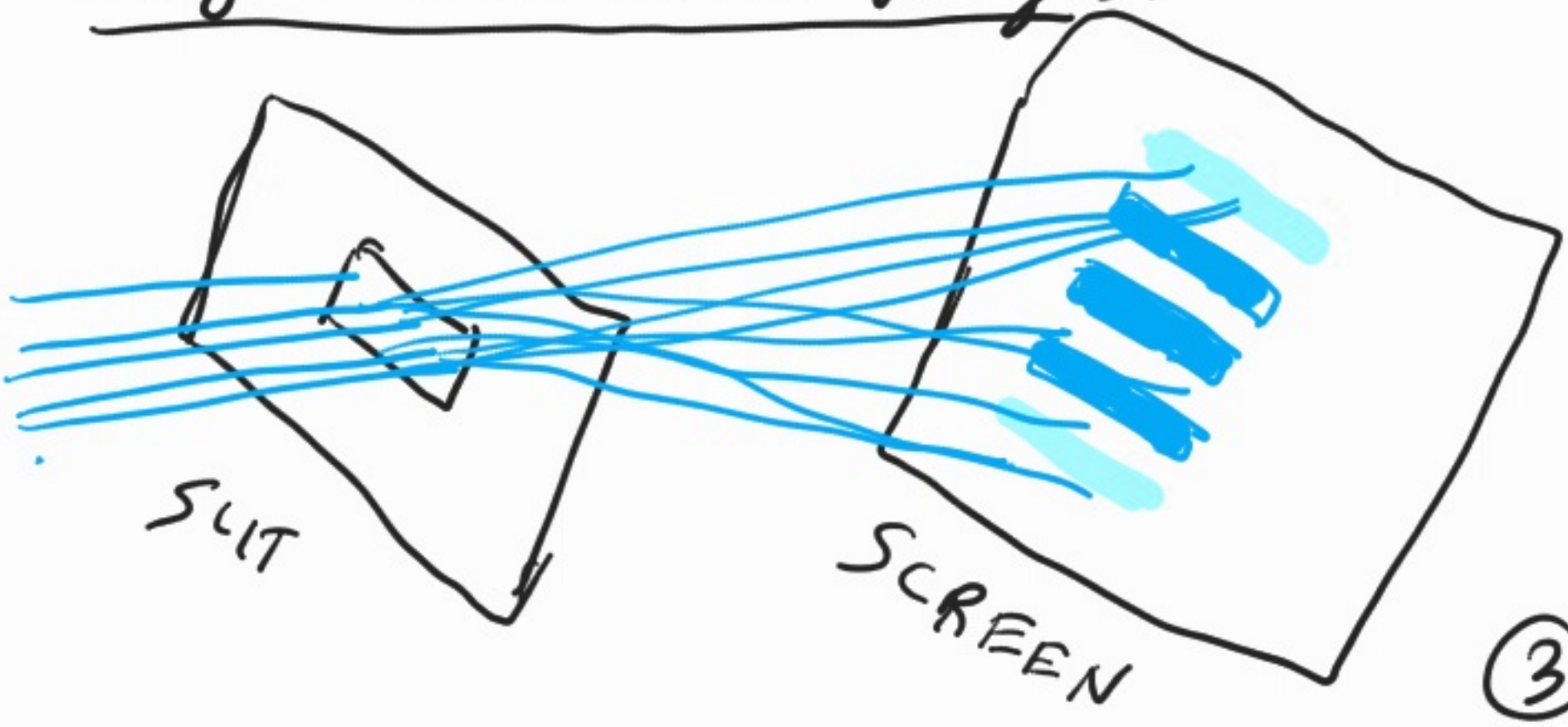


Diffraction

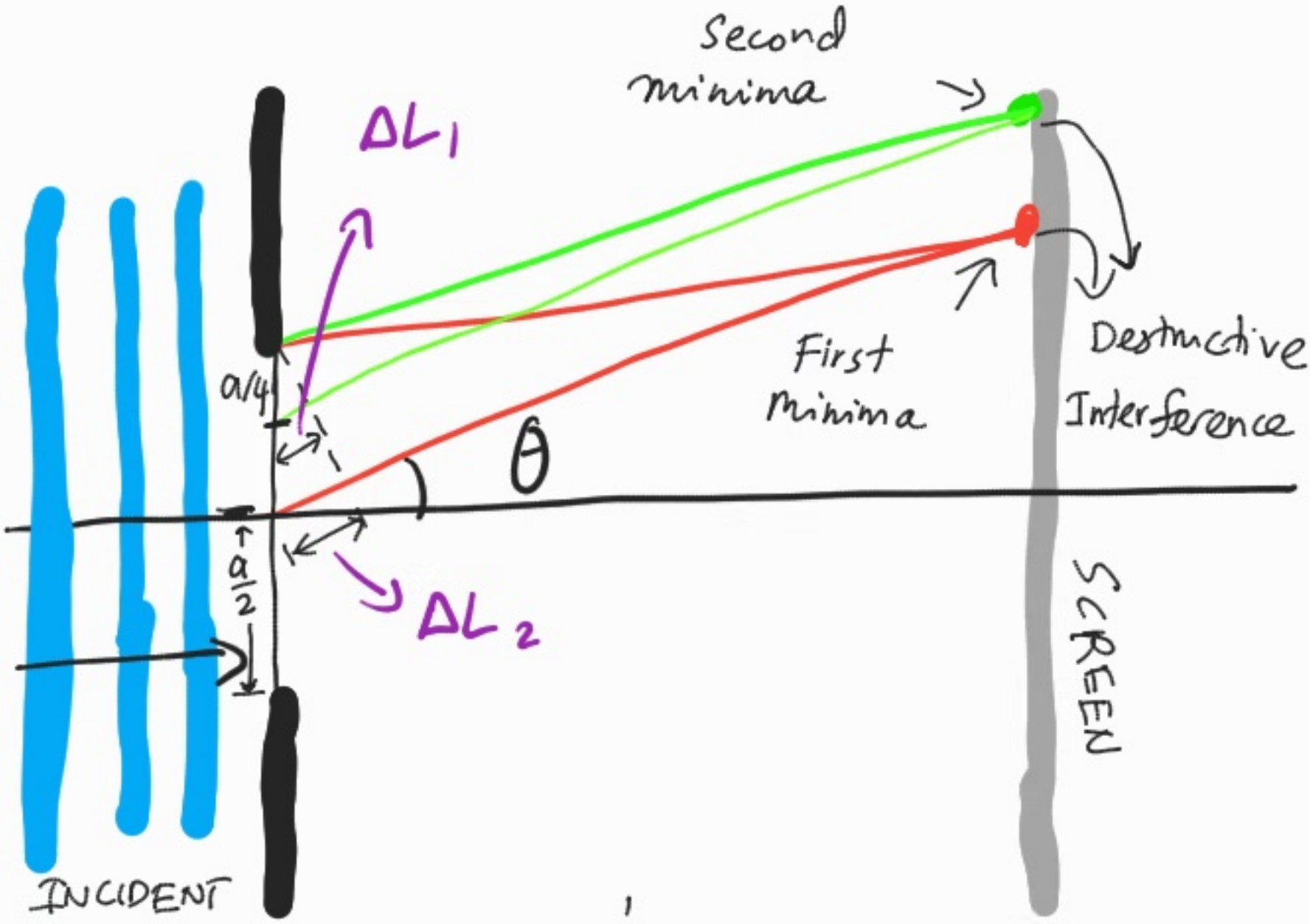
- geometric optics predicts a bright band on the screen:



- Empirically what really occurs on the screen is a set of bright and dark fringes.



Single-slit Diffraction



Path Length Difference:

$$\Delta L_1 = \frac{a}{4}\sin\theta = \frac{\lambda}{2}$$

$$\Delta L_2 = \frac{a}{2}\sin\theta = \frac{\lambda}{2}$$

First Minima:

$$a \sin\theta = \lambda$$

Second Minima:

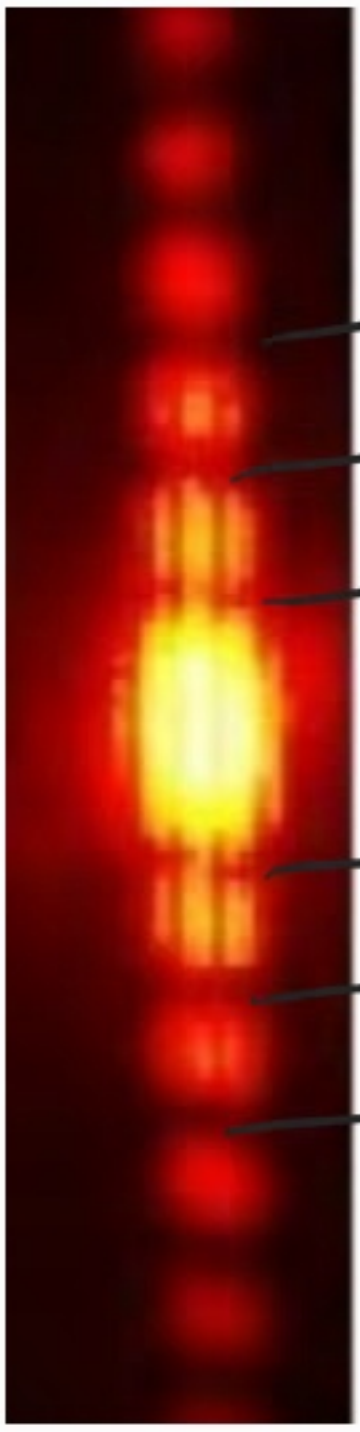
$$a \sin\theta = 2\lambda$$

All minima:

$$a \sin\theta = m\lambda$$

[DARK FRINGES]

Single-Slit Diffraction

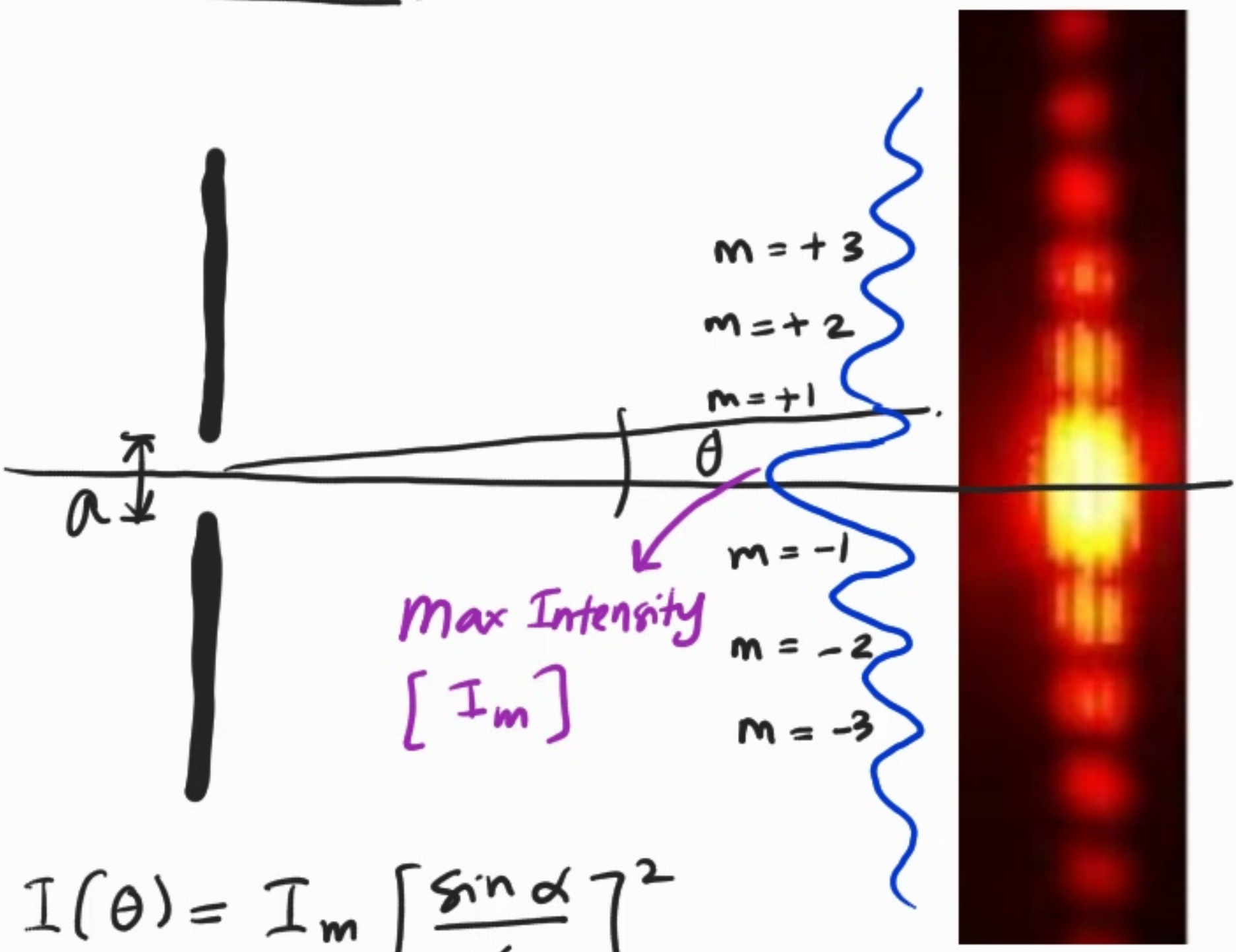


(+) : Above the central bright band

(−) : Below the central bright band.

$m = \pm 1, \pm 2, \pm 3, \ldots$

Intensity in Single-slit Diffraction.



$$I(\theta) = I_m \left[\frac{\sin\alpha}{\alpha}\right]^2$$

Intensity of Diffraction Pattern at any θ.

$$\alpha = \frac{\Delta\phi}{2} = \frac{a\pi}{\lambda}\sin\theta$$

$\Delta\phi$ ~ phase difference between top & bottom rays.

Intensity in Single-slit Diffraction

Note: $\left[\begin{matrix}\text{phase}\\ \text{Difference}\end{matrix}\right] = \frac{2\pi}{\lambda}\left[\begin{matrix}\text{Path length}\\ \text{Difference}\end{matrix}\right]$

For Minima / Dark Fringes :

$$a\sin\theta = m\lambda \Rightarrow I(\theta) = I_m\left[\frac{\sin m\pi}{m\pi}\right]^2 = 0$$

For Maxima / Bright Fringes :

$$a\sin\theta = \left[m+\frac{1}{2}\right]\lambda$$

$$I(\theta) = I_m\left[\frac{\sin\left(m+\frac{1}{2}\right)\pi}{\left(m+\frac{1}{2}\right)\pi}\right]^2$$